HW-07 — Experimental Design and System Selection

Module M09 | Chapter 11 | Weight 5% | Difficulty ⋆⋆⋆

Module: M09 Chapter: 11 Weight: 5% Due: See course schedule

Overview

Design a factorial experiment on a simulated dispatch centre, estimate main effects and interactions, apply the Rinott ranking-and-selection procedure to identify the best staffing configuration, and communicate the recommendation to a non-technical audience.

1 System

A county emergency dispatch centre handles two call types:

- **Priority 1 (P1)** — life-threatening; target answer within 10 seconds
- **Priority 2 (P2)** — non-emergency; FCFS

Baseline parameters: $\lambda_1 = 3$/hr (P1), $\lambda_2 = 12$/hr (P2), mean service times 4 min (P1) and 6 min (P2). Implement using `simpy.PriorityResource` for Pool A (P1 dispatchers).

2 Parts

2.1 Part A — 2^2 Factorial Design (30 pts)

Factors: **A** = Pool A capacity $\in \{1, 2\}$; **B** = Pool B capacity $\in \{2, 3\}$. Response: $\bar{W}_{P2}$ (mean P2 sojourn time, minutes).

Run each of the 4 treatment combinations with $r = 20$ replications. Compute the main effect of A, main effect of B, and AB interaction. Draw an interaction plot (B on x-axis, A as line parameter). Fit the linear model $\hat{W}_{P2} = \hat{\mu} + \hat{\alpha} x_A + \hat{\beta} x_B + \widehat{\alpha\beta}\, x_A x_B$ and predict the response at the centre point $(x_A, x_B) = (0, 0)$.

2.2 Part B — 2^3 Factorial (20 pts)

Add **Factor C**: $\lambda_2 \in \{10, 14\}$/hr. Run the full 2^3 design (8 treatments $\times$ 20 replications). Apply Yates' algorithm to estimate all 7 effects and interactions. Produce a Pareto chart of $|\text{effect}|$.

2.3 Part C — Ranking and Selection (30 pts)

Consider 5 staffing levels for Pool B: $c \in \{2, 3, 4, 5, 6\}$ (Pool A fixed at 1). Apply the Rinott procedure with $P^* = 0.90$, $\delta^* = 0.75$ min, pilot $n_0 = 15$. Compare to naive selection (lowest sample mean from 30 reps, repeated 100 times). How often does naive selection agree with Rinott?

2.4 Part D — Management Memo (20 pts)

$\leq$ 400 words, non-technical, addressed to the dispatch centre director. Must state: recommended configuration, expected P2 wait with uncertainty, what changes if λ_2 rises to 14/hr, and one limitation.

Grading Summary

Component	**Points**
Part A — 2^2 factorial (effects, plot, prediction)	30
Part B — 2^3 factorial (Yates, Pareto chart)	20
Part C — Rinott procedure + naive comparison	30
Part D — Management memo	20
Total	**100**

Full rubric: `rubrics/homework_rubric.pdf` | *Solutions: the instructor package (available to instructors on request).*